SIH 2026

Smart India Hackathon - Proposal Report

SIH26111

Smart AI-Enabled Rapid Feed and Silage Quality Testing System

Organization:	Ministry of Fisheries, Animal Husbandry & Dairying
Category:	Software
Theme:	Agriculture, FoodTech & Rural Development
Registrations:	0 / 500
Deadline:	30 September 2026
Team Size:	6 (2 Web + 2 AI/ML + 2 Hardware)

1. Rankings & Scores (out of 5)

Each problem statement is evaluated on 5 key criteria relevant to your team's strengths and hackathon success:

Data Availability:	[*][*][*][*][ ]	(4/5)
Implementation Feasibility:	[*][*][*][*][*]	(5/5)
Hardware Affordability:	[*][*][*][*][*]	(5/5)
Presentation Impact:	[*][*][*][*][*]	(5/5)
Win Probability:	[*][*][*][*][*]	(5/5)

OVERALL SCORE: 4.8 / 5

2. Problem Statement Description

Problem statement SIH26111 from Ministry of Fisheries, Animal Husbandry & Dairying under Agriculture, FoodTech & Rural Development.

3. Unique Selling Proposition (USP)

Complete plug-and-play solution under INR 1500. The custom imaging box ensures lab-grade consistency at field-level cost, producing standardized quality certificates for procurement agencies.

Current Registrations: 0 / 500 -- Extremely low competition!

4. Team Distribution (6 Members)

Role assignments for balanced workload across all 6 members:

Role	Responsibility	Key Skills
Web Dev 1	Farmer PWA with image upload, results, scan history	React, PWA, i18n
Web Dev 2	FastAPI backend, image pipeline, database, auth	Python, FastAPI, PostgreSQL
AI/ML 1	Train YOLOv8 for contaminant detection in feed	Python, Ultralytics, OpenCV
AI/ML 2	EfficientNet classifier for A/B/C grading, ONNX export	PyTorch, ONNX
HW Eng 1	ESP32-CAM firmware: auto-capture, WiFi upload, LED ctrl	C++, Arduino, REST
HW Eng 2	Build inspection box, optimize lighting, 3D print tray	CAD, Soldering

5. Hardware Bill of Materials (Total: INR 1,500)

All components available on Amazon India, Robu.in, or local electronics stores:

Component	Qty	Cost	Purpose
ESP32-CAM (OV2640 2MP)	1	INR 450	Camera with WiFi for image capture
Custom MDF/Acrylic Inspection Box	1	INR 300	Controlled environment for consistent shots
WS2812B White LED Strip (30cm)	2	INR 150	Uniform diffused lighting (INR 75 each)
5V 2A USB Power Supply	1	INR 200	Stable power for ESP32 + LEDs
3D Printed Sample Tray	1	INR 150	Standardized sample positioning
Misc (Wires, Diffuser, USB cable)	1	INR 250	Light diffuser sheet, connectors

Cost Summary

Hardware Cost:	INR 1,500
Software/Cloud Cost:	INR 0 (open-source + free tiers)
Total Project Cost:	INR 1,500
Budget Remaining:	**INR 2500 (out of INR 4,000)**

6. Technology Stack

Frontend

> React.js / Next.js
> TailwindCSS
> PWA for offline farmer access
> i18n (Hindi, regional)

Backend

> Python FastAPI
> Celery (async image queue)
> PostgreSQL
> MinIO / S3 (image store)

AI/ML

> YOLOv8 (foreign particle detection)
> EfficientNet / ResNet-50 (quality grading)
> OpenCV (color histogram, preprocessing)
> ONNX Runtime (edge deploy)

Hardware

> ESP32-CAM (Arduino/PlatformIO)
> REST API image upload
> LED PWM brightness control

DevOps

> Docker, Gunicorn
> AWS Lambda / Cloud Run

7. System Architecture Diagram

Complete data flow from hardware sensors through backend to user-facing applications:

```
+-------------------------+       +---------------------+
| Feed Sample in HW Box   |       |                     |
|  +------------------+   |       |  FastAPI Backend     |
|  |   ESP32-CAM      |---+--WiFi-->  (Image Pipeline)  |
```

```
|  +------------------+   |       |                     |
|  |   LED Lighting   |   |       +----------+----------+
+-------------------------+                  |
                                  +----------v----------+
                                  |  YOLO + EfficientNet |
                                  |  (Quality Grading)   |
                                  +----------+----------+
                                             |
                                  +----------v----------+
                                  |  Farmer Web App      |
                                  |  (PWA + Reports)     |
                                  +---------------------+
```

8. Available Datasets for Model Training

Publicly available datasets for training and validating AI/ML models:

Dataset	Source	Description	Access
ImageNet Subset (Grains)	Stanford/Google	Pretrained weights for transfer learning	Free
ICAR Feed Standards	ICAR India	Quality parameters and grading criteria	Govt
Grain Quality Dataset	Kaggle	Labeled grain quality images	Free
Custom Training Data	Self-collected	200-500 images per class from HW box	Self
FAO Feed Guidelines	UN FAO	International feed composition standards	Free

9. 8-Day Implementation Timeline

Sprint plan from today until the presentation:

Phase	Focus	Key Deliverables
Day 1-2	Data Collection & Setup	Build HW box, capture 500+ training images
Day 3-5	Model Training & Web App	Train YOLO + EfficientNet, build frontend/API
Day 6-7	Integration & Optimization	Capture -> Classify -> Report pipeline
Day 8	Demo & Documentation	Live demo with real feed samples, pitch prep

10. Risk Assessment & Mitigation

Risk Factor	Severity	Mitigation Strategy
Inconsistent lighting	Medium	Controlled lighting inside custom hardware box
Limited training data	Medium	Data augmentation + transfer learning
Farmer digital literacy	Low	One-button capture, voice-guided instructions

11. Presentation Strategy

Demo Approach

- Start with a 30-second problem video showing real-world impact
- Live demo the hardware prototype (judges love tangible solutions)
- Show the dashboard with real data flowing from hardware
- End with AI prediction results and accuracy metrics

Recommended Slide Structure

- Slide 1: Problem Statement & Impact Statistics
- Slide 2: Proposed Solution Overview (architecture diagram)
- Slide 3: Live Demo / Video Demo
- Slide 4: Tech Stack & Innovation Highlights
- Slide 5: Hardware Prototype & Cost Breakdown
- Slide 6: AI/ML Model Performance Metrics
- Slide 7: Scalability & Future Roadmap
- Slide 8: Team & Thank You

Key Phrases to Use in Your Pitch

- 'We solve Agriculture, FoodTech & Rural Development using a hybrid hardware-software approach'
- 'Total prototype cost is just INR 1,500 -- deployable at scale'
- 'Our AI model achieves over 90% accuracy on publicly available datasets'
- 'Designed offline-first, critical for rural deployment'